Rohan Sarkar

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LinkedIn | Portfolio

Education

Boston University, College of Engineering **Boston, MA**
Candidate for B.S. in Biomedical Engineering Expected May 2028
Kilachand Honors College | Minor: Chemistry | Concentration: Nanotechnology | GPA: 3.82/4.0
Relevant Coursework: Organic Chemistry, Biomedical Signals & Controls, Engineering Design, Statistics & Machine Learning

Research

Boston University | Teplensky Lab **Boston, MA**
Researcher April 2025 - December 2025, June 2026 - Present

- Designed and synthesized cleavable DNA dendron nanostructures to address a delivery-related limitation in target activation identified through preliminary lab data.
- Engineered nanoscale DNA platforms designed to reverse tumor immunosuppression through cGAS-STING activation, quantifying IFN-β and CXCL10 secretion by ELISA and CD80, CD86, MHC II expression in BMDCs via flow cytometry.
- Developed and optimized a glutathione-based cleavage protocol for disulfide-linked dendrons, validated by gel electrophoresis and MALDI-TOF, now used as a standard method in the lab.
- Designed anti-PD-L1-functionalized NU-1000 metal-organic framework nanoparticles to improve targeted delivery of doxorubicin to triple-negative breast cancer cells, aiming to reduce off-target toxicity, and enhance tumor cell killing.
- Encapsulated doxorubicin at 90% efficiency and achieved 78% antibody conjugation efficiency, verified using fluorescence microscopy and bicinchoninic acid (BCA) protein assays.

Work Experience

CareAI (Y Combinator S26) **Sydney, Australia**
Engineering Intern – (BU Sydney Study Abroad Program) January 2026 - June 2026

- Developed an automated document classification and extraction system utilizing optical character recognition to increase autonomous accuracy from 76% to 90%, strengthening product demo to support the company's acceptance into the YC S26 batch.
- Optimized a multi-model inference framework that generates structured evidence and decision logic for clinical data to facilitate rapid verification and error tracking.
- Designed an iterative reinforcement learning platform to model manual corrections into high-quality training data for model fine-tuning.

Office of Naval Research | Naval Information Warfare Center Atlantic **Charleston, SC**
Engineering Intern May 2024 - August 2024

- Developed a VR flight simulator using Unity and C# on the Sony Spatial Reality Display designed to engage K-12 students in government careers, enhancing their interest in STEM and public service.
- Presented project outcomes and demo to the Executive Director and Commanding Officer at Charleston Naval Base, demonstrating key functionalities and educational impact.

Projects

Predictive Heat Stroke Wearable – 1st Place Overall (1/40 teams), DS+X Hackathon October 2026

- Built a $30 wearable chest strap that predicts core body temperature from physiological and environmental sensor data, reducing prediction error by 22% over the US Army's predictive ECTemp model, validated on existing datasets containing 450 hours of data from 102 participants.

Biometrically Actuated Door Lock August 2024 - December 2024

- Developed a microcontroller-driven door locking mechanism with 94% fingerprint validation to improve accessibility for elderly and disabled users, eliminating the need for physical keys and manual turning movements.

Awards

- DS+X Hackathon: 1st Place Overall and 1st Place, Health Track October 2026
- Undergraduate Research Opportunities Program: competitive, funded faculty-mentored research June 2026, October 2026
- Dr. Murray Brockman Leadership Award: awarded for leadership, character, and academic excellence May 2024
- South Carolina Junior Academy of Science Research Award: 3rd place Chemistry Division March 2024

Skills

- **Wet Lab:** Cell Culture, Primary & Bone Marrow Cell Isolation and Differentiation, Confocal Microscopy, Flow Cytometry, ELISA, HPLC, MALDI-TOF, ESI-LC-MS, DNA Synthesis, BCA, Gel Electrophoresis, Aseptic Technique
- **Programming:** MATLAB, C, C++, C#, Java, Python, SQL, Arduino, R
- **CAD:** SolidWorks, Fusion360, Onshape, 3D Printing
- **Technical**: LLM, NLP, Microsoft 365, Google Workspace
- **Languages:** English (native), Bengali (native)